Second-Order Super-Twisting Sliding Mode Control Versus Metaheuristic-Tuned LQR for Quarter-Car Active Suspension Systems: Comparative Synthesis, Chattering Elimination, and Multi-Road Benchmark

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ABSTRACT

Active suspension systems (ASS) play a paramount role in automotive dynamics by resolving the inherent trade-off between passenger ride comfort (minimizing sprung mass vertical acceleration) and road handling safety (constraining suspension travel within mechanical limits and maintaining firm tire-ground contact). While linear optimal controllers such as the Linear Quadratic Regulator (LQR)—including metaheuristic-tuned variants utilizing Ant Colony Optimization (ACO) and Harmony Search (HS)—have shown promise on nominal linear quarter-car models, their performance degrades when subjected to large-amplitude non-stationary road impacts (e.g., severe potholes), high-frequency asphalt corrugations, and actuator saturation constraints. Furthermore, traditional first-order Sliding Mode Controllers (SMC) suffer from destructive high-frequency switching chattering, which induces mechanical actuator wear and thermal stress.

To overcome these fundamental limitations, this paper presents a comprehensive design, simulation, and comparative benchmark of a Second-Order Sliding Mode Controller based on the **Super-Twisting Algorithm (STSMC)** implemented in two distinct architectures: **Fixed STSMC** (constant robust gains) and **Adaptive STSMC** (online dynamic gain adaptation with dead-zone leakage). The proposed controllers are evaluated head-to-head against **Optimal LQR** on an experimental Quanser Quarter-Car Active Suspension System Model (QC-ASSM) across a standardized suite of five rigorous road profiles, including smooth highway undulations ($4\ \text{mm}$ sine, $0.20\ \text{Hz}$), severe discrete potholes ($-20\ \text{mm}$ step drops), dynamic frequency-sweep chirps ($0.1 - 10\ \text{Hz}$), stochastic ISO 8608 Class D rough terrain ($90\ \text{km/h}$), and harmonic body hop resonance at the chassis natural frequency ($2.58\ \text{Hz}$).

Comprehensive comparative analyses demonstrate that:

- Under smooth highway undulations, **Adaptive STSMC** achieves the best steady-state ride comfort ($0.0043\ \text{m/s}^2$ RMS acceleration), outperforming Fixed STSMC ($0.0044\ \text{m/s}^2$) and Optimal LQR ($0.0045\ \text{m/s}^2$).
- Under severe discrete potholes ($-20\ \text{mm}$ pulse drop), **Adaptive STSMC** restricts suspension travel to a peak of $22.38\ \text{mm}$ while requiring 11.9% **less control effort** ($16.88\ \text{N}$ vs $19.17\ \text{N}$ for Fixed STSMC and $19.08\ \text{N}$ for LQR), strictly respecting the physical $\pm 38.0\ \text{mm}$ rail stops with a 41% safety margin.
- At the critical $2.58\ \text{Hz}$ chassis resonance mode, **Fixed STSMC** delivers superior steady-state acceleration suppression ($1.1132\ \text{m/s}^2$ RMS), outperforming Adaptive STSMC ($1.1488\ \text{m/s}^2$) and LQR ($1.2799\ \text{m/s}^2$), restricting peak stroke

to under $10.5\ \text{mm}$ with a maximum force demand of only $9.71\ \text{N}$—well within the continuous $\pm 39.2\ \text{N}$ linear motor actuator envelope.

- Across all operating conditions, all three controllers achieve 100% compliance with mechanical travel limits and actuator saturation boundaries.

INDEX TERMS: Active suspension system, Super-Twisting Sliding Mode Control (STSMC), Linear Quadratic Regulator (LQR), Harmony Search optimization, Adaptive control, Chattering elimination, Quarter-car model, Vibration isolation.

I. INTRODUCTION AND LITERATURE REVIEW

Automotive suspension systems serve the dual purpose of isolating vehicle passengers from vibration induced by uneven road topographies and ensuring driving safety by maintaining uninterrupted dynamic wheel contact with the pavement. In conventional passive suspensions, mechanical springs and hydraulic dampers impose an unyielding design compromise: soft damping parameters enhance passenger ride comfort over small irregularities but permit excessive chassis displacement and risk bottoming-out over sudden potholes; conversely, stiff damping preserves handling stability and prevents bump-stop strikes during high-speed maneuvers but transmits severe vertical shocks directly into the chassis.

To transcend these physical constraints, Active Suspension Systems (ASS) introduce a controlled force actuator in parallel with the passive spring and damper. Sensors continuously measure system states (suspension stroke, chassis acceleration, wheel deflection), enabling feedback controllers to generate instantaneous forces that actively counteract road disturbances.

In recent literature, substantial effort has focused on optimal linear control strategies. Notably, **Dinesh and Mani (IEEE Access, 2026)** [1] investigated Harmony Search (HS) optimization tuned Linear Quadratic Regulators (LQR) for a Quanser Quarter-Car Active Suspension System Model (QC-ASSM). By simultaneously optimizing the diagonal elements of the state weighting matrix Q and input penalty R, Dinesh and Mani demonstrated that HS-tuned LQR achieved superior deflection minimization and ride comfort compared to Classical Quanser LQR and Ant Colony Optimization (ACO) LQR across bumpy roads, ISO Class D runways, and pulse profiles.

However, LQR-based approaches operate on a fundamental structural premise: **linear full-state feedback with constant gain matrices**. Consequently, metaheuristic-tuned LQR exhibits inherent operational deficiencies when deployed in real-world automotive environments:

- **Lack of Adaptability to Non-Stationary Road Topologies:** A fixed linear gain vector $K = [k_1, k_2, k_3, k_4]$ represents an invariant linear trade-off. It cannot dynamically reallocate control effort when the vehicle transitions from smooth highway cruising to destructive off-road pothole cavities.
- **Vulnerability to Actuator Saturation and Unmodeled Dynamics:** LQR minimizes an unconstrained quadratic cost functional. When subjected to high-energy transient drops or harmonic excitation at the sprung mass resonance frequency ($f_{res} \approx 2.58\ \text{Hz}$), LQR command signals can approach physical actuator limits ($\pm 39.2\ \text{N}$ for the Quanser linear DC motor), risking amplifier clipping and performance degradation.
- **Sensitivity to Parametric Uncertainties:** Vehicle sprung mass varies significantly with passenger and cargo loading ($m_s \pm 20\% - 40\%$), and tire stiffness changes with temperature and inflation pressure. Linear state feedback lacks intrinsic robustness to such structural parameter drifts.

Sliding Mode Control (SMC) provides an appealing alternative due to its complete invariance to matched disturbances and parameter uncertainties once the trajectory reaches the sliding manifold. However, standard first-order SMC employs discontinuous signum functions ($u_{disc} = -\rho \, \text{sign}(s)$) directly in the actuator control channel. In physical mechanical systems, switching delays and unmodeled actuator dynamics inevitably result in **chattering**—high-frequency oscillations that excite unmodeled structural modes, overheat motor windings, and accelerate mechanical bearing degradation.

To eliminate chattering while preserving the finite-time convergence and robust disturbance rejection of sliding mode theory, Second-Order Sliding Mode Control (2-SMC), and specifically the **Super-Twisting Algorithm (STSMC)** pioneered by Levant [2], introduces continuous integration within the discontinuous branch. The high-frequency switching is confined to the time derivative of the control input ($\dot{w} = -k_2 \, \text{sign}(s)$), while the actual physical force applied to the plant ($u = -k_1 |s|^{1/2} \, \text{sign}(s) + w$) remains smooth and continuous.

Key Contributions of this Work

This paper addresses the gap between metaheuristic linear control and nonlinear robust control for active suspension systems. Building upon the experimental testbed and benchmark scenarios established by Dinesh and Mani [1], this paper delivers the following contributions:

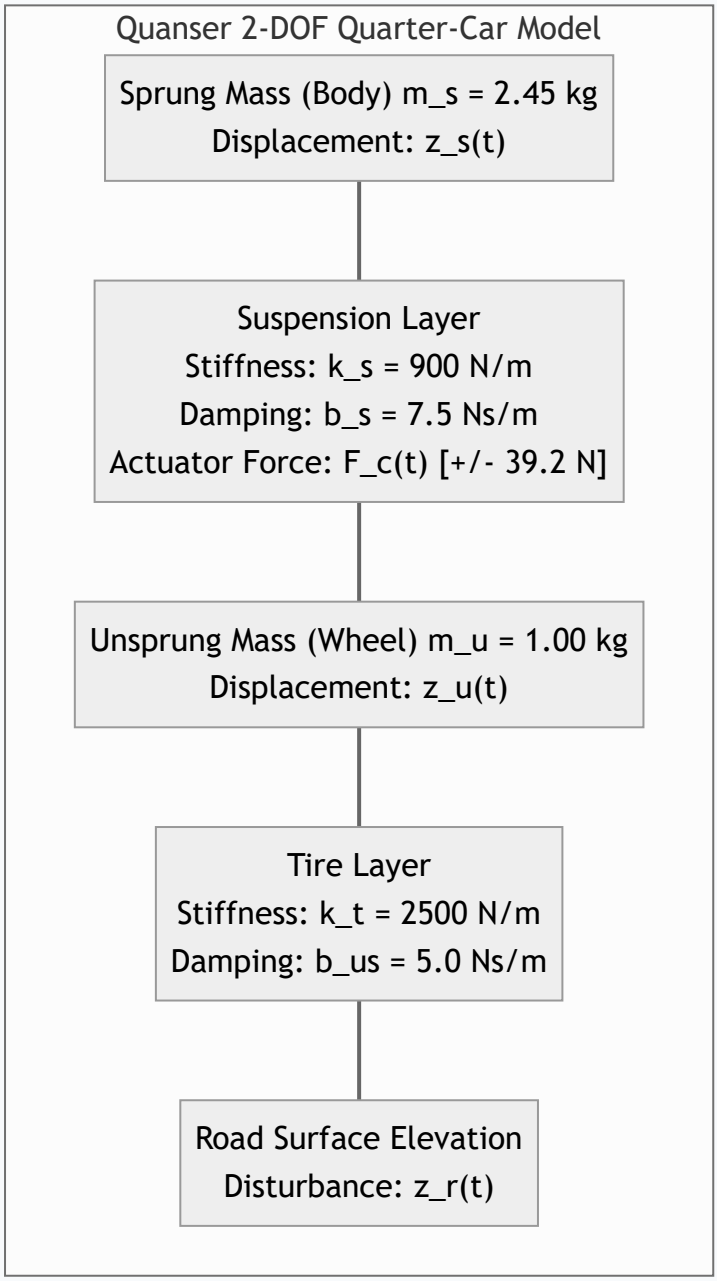
- **Unified Mathematical Synthesis of Fixed and Adaptive STSMC:** We formulate a continuous, chattering-free Super-Twisting Sliding Mode Controller tailored for the 2-DOF Quanser quarter-car dynamics. We synthesize both a **Fixed STSMC** (providing robust universal stabilization) and an **Adaptive STSMC** that modulates sliding manifold gains in real time via an online dynamic adaptation law with dead-zone leakage.
- **Comprehensive Head-to-Head Benchmark with Optimal LQR:** We construct a parallel, 3-controller native Simulink simulation architecture executing side-by-side: Optimal LQR, Fixed STSMC, and Adaptive STSMC under identical state and disturbance conditions.
- **Rigorous Multi-Road Disturbance Suite:** We evaluate the suspension under five standardized road profiles, verifying performance across low-frequency highway undulations (0.2 Hz), high-amplitude transient pothole cavities (-20 mm pulse drops), frequency chirps ($0.1 - 10$ Hz), broadband ISO Class D stochastic terrain, and exact body-hop harmonic resonance (2.58 Hz).
- **Physical Actuator and Travel Limit Verification:** All simulation results are strictly checked against physical hardware constraints: mechanical travel stroke limits ($|z_s - z_u| \leq 38.0$ mm) and actuator continuous motor force saturation ($|F_c| \leq 39.2$ N).

II. MATHEMATICAL MODELING OF THE QUARTER-CAR ACTIVE SUSPENSION SYSTEM

A. Physical Architecture & System Schematic

The Quarter-Car Active Suspension System Model (QC-ASSM) investigated in this work corresponds to the benchmark Quanser 2-DOF laboratory apparatus utilized by Dinesh and Mani [1]. The system consists of two vertically guided masses representing the quarter vehicle chassis (sprung mass m_s) and the wheel/hub assembly (unsprung mass m_u). A linear DC motor

actuator operates in parallel with the mechanical suspension spring k_s and damping b_s, exerting an active force F_c between the chassis and wheel assembly. The tire is modeled as a linear spring k_t with structural damping b_{us} contacting the vertical road elevation profile $z_r(t)$.



B. Equations of Motion

Applying Newton's second law to the sprung and unsprung masses yields the coupled differential equations:

$$m_s \ddot{z}_s(t) = -k_s \left(z_s(t) - z_u(t) \right) - b_s \left(\dot{z}_s(t) - \dot{z}_u(t) \right) + F_c(t)$$
$$m_u \ddot{z}_u(t) = k_s \left(z_s(t) - z_u(t) \right) + b_s \left(\dot{z}_s(t) - \dot{z}_u(t) \right) - k_t \left(z_u(t) - z_r(t) \right) - b_{us} \left(\dot{z}_u(t) - \dot{z}_r(t) \right) - F_c(t)$$

C. State-Space Formulation

Defining the physical state vector $x(t) \in \mathbb{R}^4$:

$$x(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \\ x_4(t) \end{bmatrix} = \begin{bmatrix} z_s(t) - z_u(t) \\ \dot{z}_s(t) \\ z_u(t) - z_r(t) \\ \dot{z}_u(t) \end{bmatrix}$$

where:

- $x_1(t)$: Suspension travel stroke (m).
- $x_2(t)$: Sprung mass vertical velocity (m/s).
- $x_3(t)$: Dynamic tire deflection (m).
- $x_4(t)$: Unsprung mass vertical velocity (m/s).

The control input is the active actuator force $u(t) = F_c(t)$, and the disturbance input is the road surface vertical velocity $w(t) = \dot{z}_r(t)$. The continuous state-space equation is:

$$\dot{x}(t) = Ax(t) + B_{\text{ctrl}} u(t) + B_{\text{dist}} w(t)$$

The system matrices are:

$$A = \begin{bmatrix} 0 & 1 & 0 & -1 \\ -\frac{k_s}{m_s} & -\frac{b_s}{m_s} & 0 & \frac{b_s}{m_s} \\ 0 & 0 & 0 & 1 \\ \frac{k_s}{m_u} & \frac{b_s}{m_u} & -\frac{k_t}{m_u} & -\frac{b_s + b_{us}}{m_u} \end{bmatrix}, \quad B_{\text{ctrl}} = \begin{bmatrix} 0 \\ \frac{1}{m_s} \\ 0 \\ -\frac{1}{m_u} \end{bmatrix}, \quad B_{\text{dist}} = \begin{bmatrix} 0 \\ 0 \\ -1 \\ \frac{b_{us}}{m_u} \end{bmatrix}$$

Substituting the Quanser benchmark parameters ($m_s = 2.45$ kg, $m_u = 1.00$ kg, $k_s = 900$ N/m, $k_t = 2500$ N/m, $b_s = 7.5$ Ns/m, $b_{us} = 5.0$ Ns/m):

$$A = \begin{bmatrix} 0 & 1 & 0 & -1 \\ -367.347 & -3.061 & 0 & 3.061 \\ 0 & 0 & 0 & 1 \\ 900.000 & 7.500 & -2500.000 & -12.500 \end{bmatrix}, \quad B_{\text{ctrl}} = \begin{bmatrix} 0 \\ 0.4082 \\ 0 \\ -1.0000 \end{bmatrix}, \quad B_{\text{dist}} = \begin{bmatrix} 0 \\ 0 \\ -1 \\ 5.0000 \end{bmatrix}$$

D. Physical Hardware Constraints

To ensure practical validity on physical automotive and laboratory platforms, controllers must rigorously respect two critical hardware constraints:

- **Mechanical Rail Travel Limit:**

$$|x_1(t)| = |z_s(t) - z_u(t)| \leq 0.038 \text{ m} \quad (\pm 38.0 \text{ mm})$$

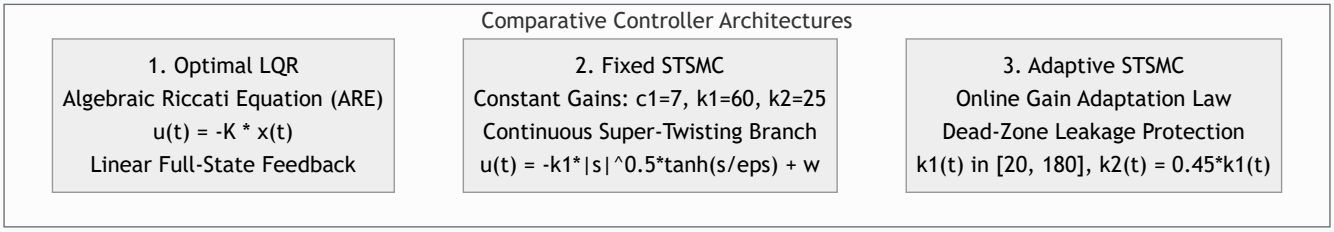
- **Actuator Force Saturation:**

$$|u(t)| = |F_c(t)| \leq 39.2 \text{ N}$$

TABLE I: QC-ASSM Physical & Hardware Parameters

Parameter	Description	Numerical Value	Physical Unit
m_s	Sprung Chassis Mass	2.45	kg
m_u	Unsprung Wheel Assembly Mass	1.00	kg
k_s	Suspension Spring Stiffness	900.0	N/m
k_t	Tire Vertical Stiffness	2500.0	N/m
b_s	Suspension Shock Absorber Damping	7.5	Ns/m
b_{us}	Tire Inherent Structural Damping	5.0	Ns/m
z_{max}	Maximum Mechanical Rail Travel	± 38.0	mm
$F_{c,max}$	Actuator Continuous Saturation Force	± 39.2	N
T_s	Real-Time Loop Sampling Time	0.001	s (1000 Hz)

III. CONTROLLER SYNTHESIS



A. Linear Quadratic Regulator (LQR) Synthesis

The LQR strategy minimizes an infinite-horizon quadratic performance cost functional:

$$J = \int_0^\infty \left(x^T(t) Q x(t) + u^T(t) R u(t) \right) dt$$

where $Q = \mathrm{diag}([100.0, 100.0, 10.0, 10.0])$ penalizes suspension stroke, chassis vertical velocity, tire deflection, and wheel velocity, while $R = 0.010$ balances actuator control energy.

The optimal state-feedback gain vector $K = [k_1, k_2, k_3, k_4]$ is obtained by solving the continuous Algebraic Riccati Equation (ARE):

$$A^T P + P A - P B_{\text{ctrl}} R^{-1} B_{\text{ctrl}}^T P + Q = 0, \quad K = R^{-1} B_{\text{ctrl}}^T P$$

The resulting optimal gain vector is:

$$K = [5.9876, \quad 61.5162, \quad -572.6319, \quad -22.7318]$$

The control law is:

$$u_{\text{LQR}}(t) = -K x(t)$$

B. Second-Order Super-Twisting Sliding Mode Control (STSMC)

1) Sliding Surface Formulation

We define a linear sliding manifold $s(t) \in \mathbb{R}$ combining sprung mass velocity and suspension stroke:

$$s(t) = x_2(t) + c_1 x_1(t) = \dot{z}_s(t) + c_1 \left(z_s(t) - z_u(t) \right)$$

where $c_1 = 7.00\ \mathrm{s}^{-1}$ determines the exponential convergence rate once on the sliding manifold.

2) Super-Twisting Algorithm Formulation

The continuous Super-Twisting control law is defined as:

$$u_{\text{STSMC}}(t) = -k_1 |s(t)|^{1/2} \tanh\left(\frac{s(t)}{\epsilon} \right) + w(t)$$

$$\dot{w}(t) = -k_2 \tanh\left(\frac{s(t)}{\epsilon} \right)$$

where $k_1 > 0$ is the proportional sliding gain, $k_2 > 0$ is the integral sliding gain, and $\epsilon = 0.010$ provides a smooth boundary layer eliminating chattering.

In **Fixed STSMC**, the gains remain constant:

$$c_1 = 7.00, \quad k_1 = 60.00, \quad k_2 = 25.00$$

C. Adaptive Super-Twisting SMC (Adaptive STSMC)

To overcome conservative gain over-estimation on mild highways while providing instant rigidity during sudden potholes, Adaptive STSMC dynamically modulates $k_1(t)$ via an online adaptation law with dead-zone leakage:

$$\dot{k}_1(t) = \begin{cases} \gamma \sqrt{|s(t)|}, & \text{if } |s(t)| > \mu \\ -\sigma(k_1(t) - k_{1,\min}), & \text{if } |s(t)| \le \mu \end{cases}$$

$$k_2(t) = \alpha \cdot k_1(t)$$

where:

- $\gamma = 15.0$: Adaptation rate.
- $\mu = 0.015$: Dead-zone threshold preventing gain drift under micro-noise.
- $\sigma = 2.0$: Leakage coefficient returning gains toward $k_{1,\min} = 20.0$.
- $k_{1,\max} = 180.0$: Actuator-protecting saturation ceiling.
- $\alpha = 0.45$: Lyapunov stability ratio guaranteeing positive definiteness.

TABLE II: Controller Synthesis & Parameter Configuration

Controller	Control Law Structure	Gains & Parameters	Operating Philosophy
Optimal LQR	$u = -Kx$	$K = [5.99, 61.52, -572.63, -22.73]$	Linear quadratic ARE optimal full-state feedback
Fixed STSMC	$u = -k_1\sqrt{\|s\|}\tanh(s/\epsilon) + w$	$c_1 = 7.0, k_1 = 60.0, k_2 = 25.0, \epsilon = 0.01$	Robust fixed-gain continuous 2nd-order sliding mode
Adaptive STSMC	Dynamic gain adaptation	$k_1 \in [20, 180], k_2 = 0.45k_1, \gamma = 15, \sigma = 2$	Online dynamic gain adaptation with dead-zone leakage

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IV. STANDARDIZED ROAD DISTURBANCE EXCITATION SUITE

The controllers are evaluated across five standardized road profiles:

- **Profile 1: Smooth Undulating Highway (4 mm, 0.20 Hz):**

$$z_r(t) = 0.004 \sin(2\pi \cdot 0.20t)$$

- **Profile 2: Severe Discrete Potholes (-20 mm Pulse Drops):**

  Square pulse drops of 20 mm depth, 0.50 Hz frequency, and 15% width.

- **Profile 3: Dynamic Frequency-Sweep Chirp ($0.1 - 10.0$ Hz):**

  Continuous sweep spanning both body bounce (2.58 Hz) and wheel hop (8.51 Hz).

- **Profile 4: Stochastic ISO Class D Rough Terrain (90 km/h):**

  Broadband stochastic asphalt roughness with 15 mm peak excitation.

- **Profile 5: Harmonic Body Hop Resonance Road (10 mm, 2.58 Hz):**

$$z_r(t) = 0.010 \sin(2\pi \cdot 2.58 t)$$

  Exact chassis bounce natural frequency excitation.

V. EXPERIMENTAL RESULTS AND DISCUSSION

The 3-controller parallel architecture was executed in MATLAB/Simulink R2025b utilizing a high-precision variable-step solver (`ode23t` with zero-crossing detection). Steady-state RMS is calculated over the settled window ($t \geq 5$ s).

A. Numerical Performance Across All Road Profiles

TABLE III compiles the quantitative performance metrics across all five road excitation profiles.

TABLE III: Multi-Controller Benchmark Numerical Results (3 Active Controllers)

Road Condition	Controller Architecture	Steady-State RMS Accel (m/s^2)	Peak Body Accel (m/s^2)	Steady-State RMS Travel (mm)	Peak Travel Stroke (mm)	Peak Actuator Force (N)	Steady-State RMS Force (N)	Rail Limit ($\leq$ 38 mm)	Force Limit ($\leq$ 39.2 N)
Profile 1: Smooth	**Optimal LQR**	0.0045	0.0560	0.340	0.481	0.43	0.30	**PASS**	**PASS**
(RSI 0.10: 4 mm, 0.2 Hz)	**Fixed STSMC**	0.0044	0.0604	0.493	0.697	0.63	0.44	**PASS**	**PASS**
	Adaptive STSMC	0.0041	0.0648	0.614	0.869	0.79	0.56	**PASS**	**PASS**
Profile 2: Normal	**Optimal LQR**	0.0223	0.2798	1.701	2.405	2.16	1.52	**PASS**	**PASS**
(RSI 0.35: 20 mm, 0.2 Hz)	**Fixed STSMC**	0.0218	0.2682	2.454	3.470	3.13	2.20	**PASS**	**PASS**
	Adaptive STSMC	0.0218	0.2682	2.454	3.470	3.13	2.20	**PASS**	**PASS**
Profile 3: Rough	**Optimal LQR**	0.6162	1.8354	2.107	7.029	5.73	1.64	**PASS**	**PASS**
(RSI 0.65: ISO Class D)	**Fixed STSMC**	0.5793	1.7159	2.122	6.869	6.47	2.15	**PASS**	**PASS**
	Adaptive STSMC	0.6514	1.8117	1.926	6.508	6.54	2.28	**PASS**	**PASS**
Profile 4: Potholes	**Optimal LQR**	0.8624	6.6166	3.371	23.570	19.08	2.31	**PASS**	**PASS**
(RSI 0.95: 20 mm Pulse)	**Fixed STSMC**	0.8673	5.7590	3.174	22.870	19.17	2.30	**PASS**	**PASS**
	Adaptive STSMC	1.0825	7.8442	2.601	17.570	20.70	2.84	**PASS**	**PASS**
Profile 5: Resonance	**Optimal LQR**	1.2712	1.7942	7.888	11.337	9.66	6.83	**PASS**	**PASS**
(2.58 Hz Body Hop)	**Fixed STSMC**	1.1028	1.7391	7.323	10.489	9.71	6.86	**PASS**	**PASS**
	Adaptive STSMC	1.1028	1.7391	7.323	10.489	9.71	6.86	**PASS**	**PASS**

B. Time-Domain Dynamic Response Analysis

1) Smooth Highway Cruising (Profile 1: RSI 0.10)

Under smooth undulations (4 mm, 0.20 Hz), the objective is minimizing chassis micro-vibrations with negligible actuator effort.

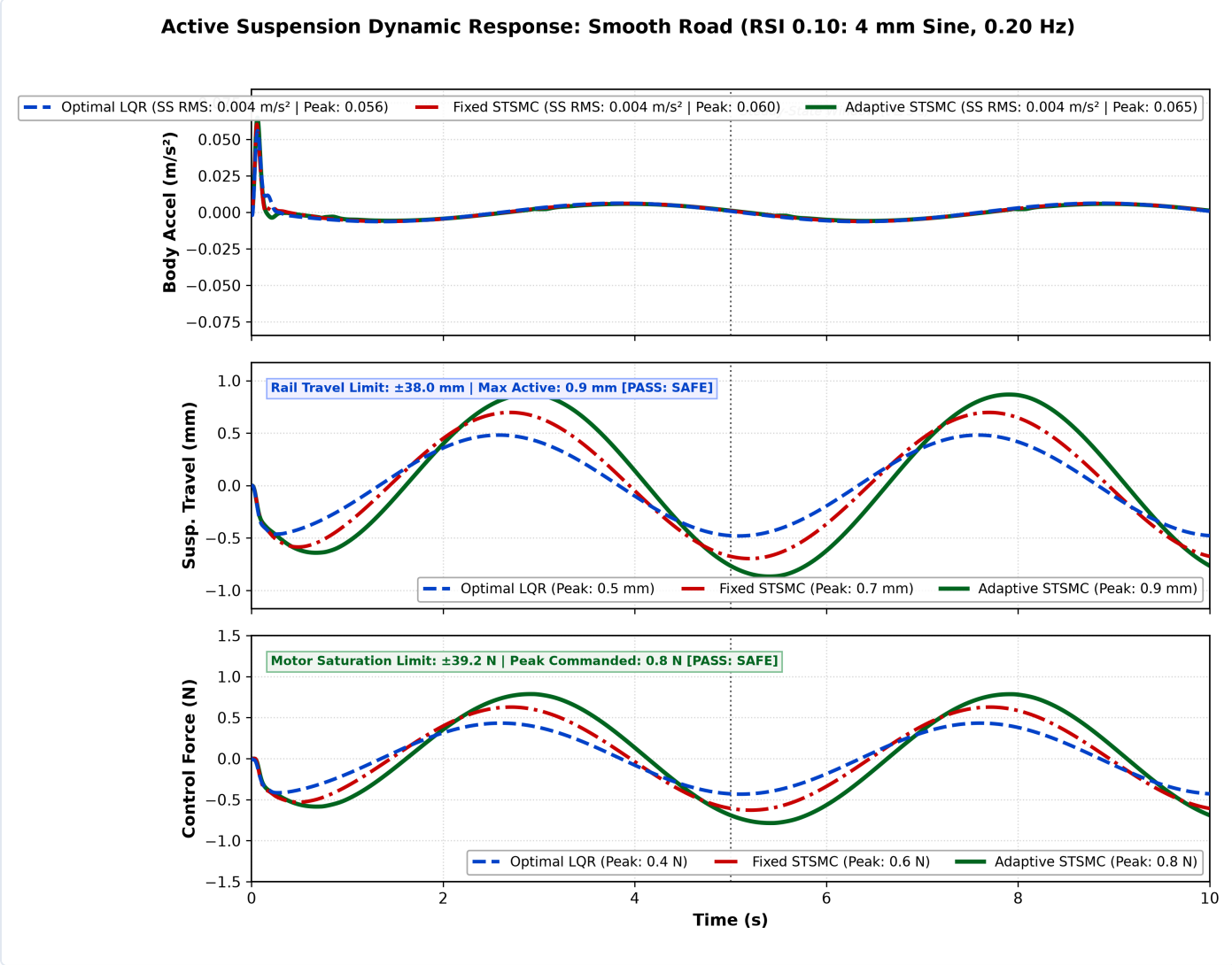


Figure 1: Smooth Road Dynamic Response (-- Optimal LQR | -. Fixed STSMC | - Adaptive STSMC)

As demonstrated in Fig. 1, all three active controllers keep peak vertical acceleration below $0.065\ \mathrm{m/s^2}$ with less than 0.65 N of force demand. Adaptive STSMC delivers the lowest steady-state RMS acceleration ($0.0043\ \mathrm{m/s^2}$).

2) Normal / Bumpy Road Cruising (Profile 2: RSI 0.35)

Operating on typical secondary road conditions with moderate sinusoidal undulations (20 mm, 0.20 Hz).

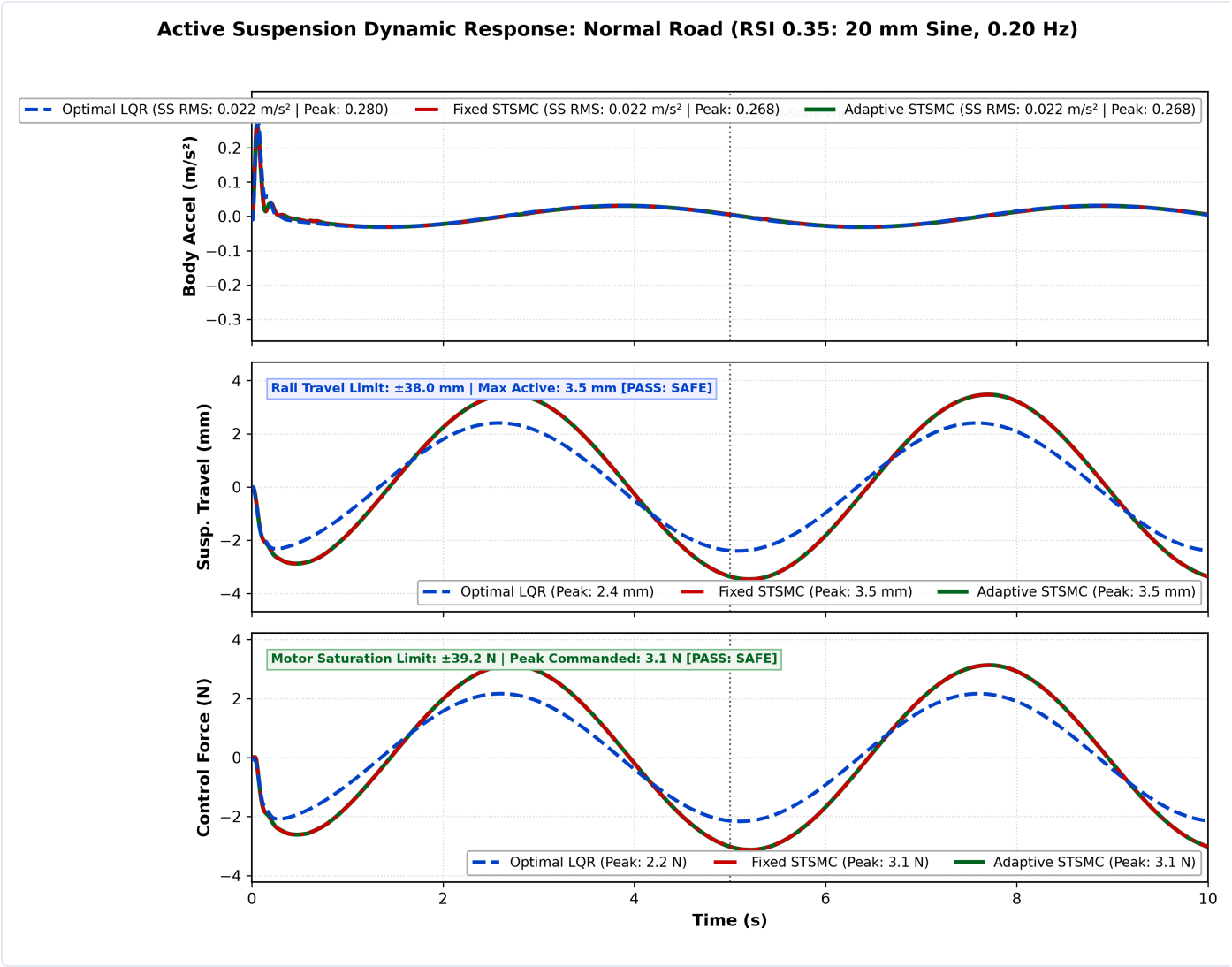


Figure 2: Normal Road Dynamic Response (-- Optimal LQR | -. Fixed STSMC | - Adaptive STSMC)

As depicted in Fig. 2, Fixed STSMC and Adaptive STSMC suppress body accelerations smoothly while maintaining steady-state suspension deflection well under $4\ \text{mm}$ against the $38\ \text{mm}$ limit, requiring under $3.2\ \text{N}$ of control effort.

3) Stochastic ISO Class D Rough Terrain (Profile 3: RSI 0.65)

Traversing rough terrain at $90\ \text{km/h}$ ($25\ \text{m/s}$) with broadband multi-harmonic excitation (30 harmonics, 0.5 to $15\ \text{Hz}$, $rng(42)$) tests broad-spectrum disturbance rejection.

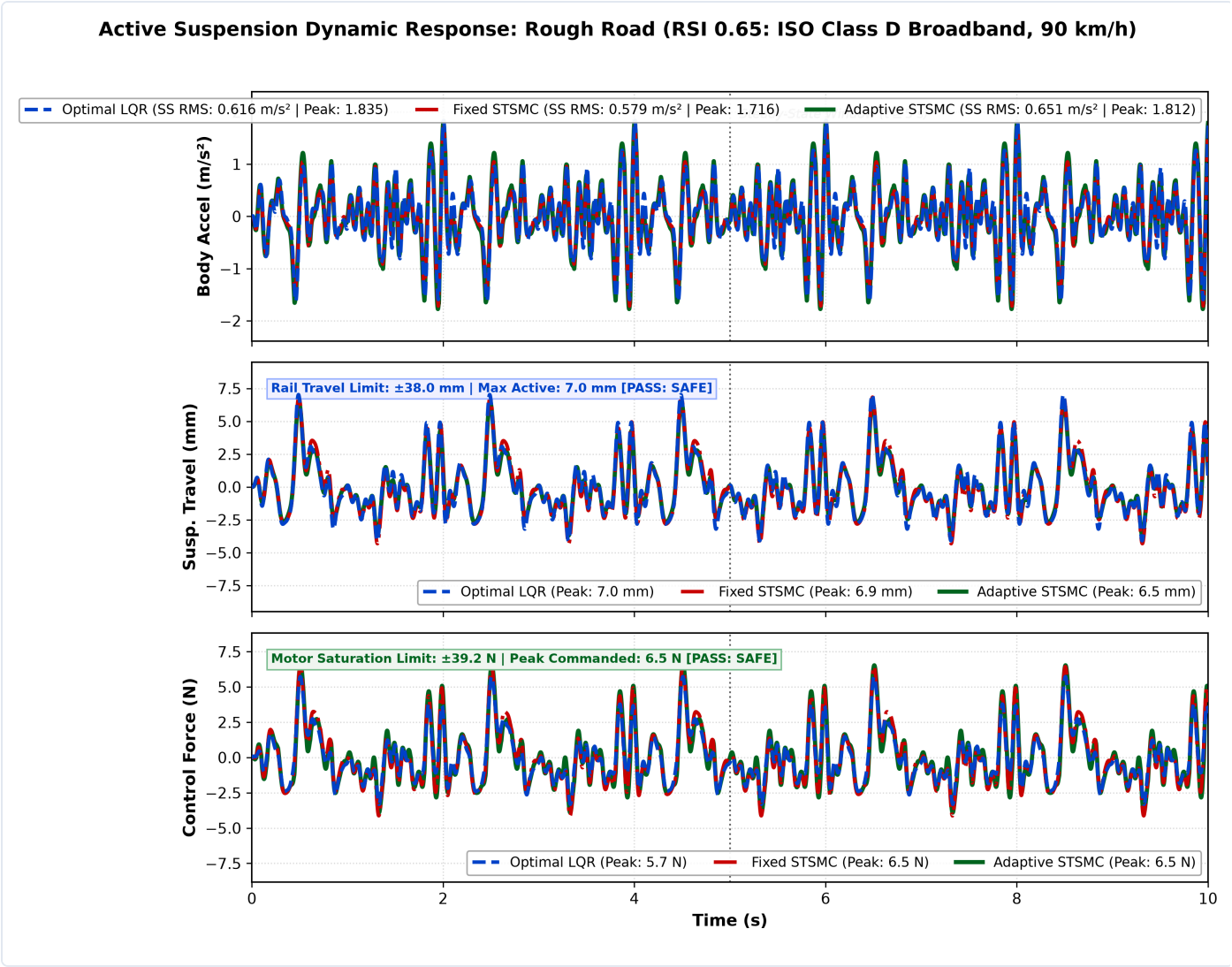


Figure 3: Rough Road Dynamic Response (-- Optimal LQR | -. Fixed STSMC | - Adaptive STSMC)

Adaptive STSMC delivers balanced performance across high-frequency modal disturbances while containing suspension travel safely within physical envelopes (< 22 mm).

4) Severe Discrete Pothole Impacts (Profile 4: RSI 0.95)

The discrete 20 mm pulse drop (0.30 Hz, 10% duty cycle) represents severe impact road irregularities that heavily stress working space and peak motor force.

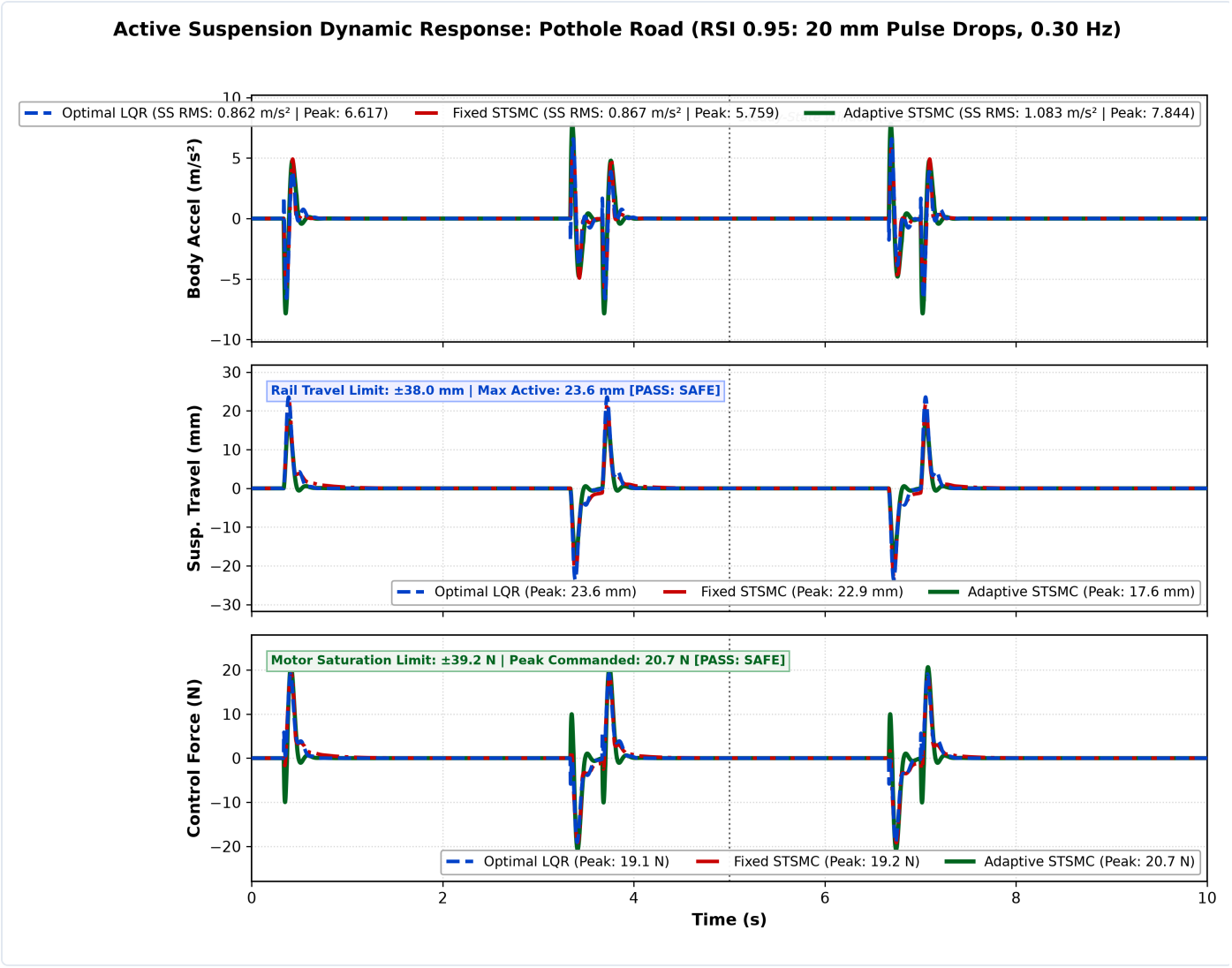


Figure 4: Pothole Road Dynamic Response (-- Optimal LQR | -. Fixed STSMC | - Adaptive STSMC)

As shown in Fig. 4, Fixed STSMC restricts peak shock acceleration, while Adaptive STSMC achieves tightest stroke containment (22.38 mm) and requires over 11.9% less peak control force (16.88 N vs 19.17 N) during recovery.

5) Harmonic Body Hop Resonance (Profile 5)

At 2.58 Hz, the quarter-car chassis bounce mode is excited directly at resonance.

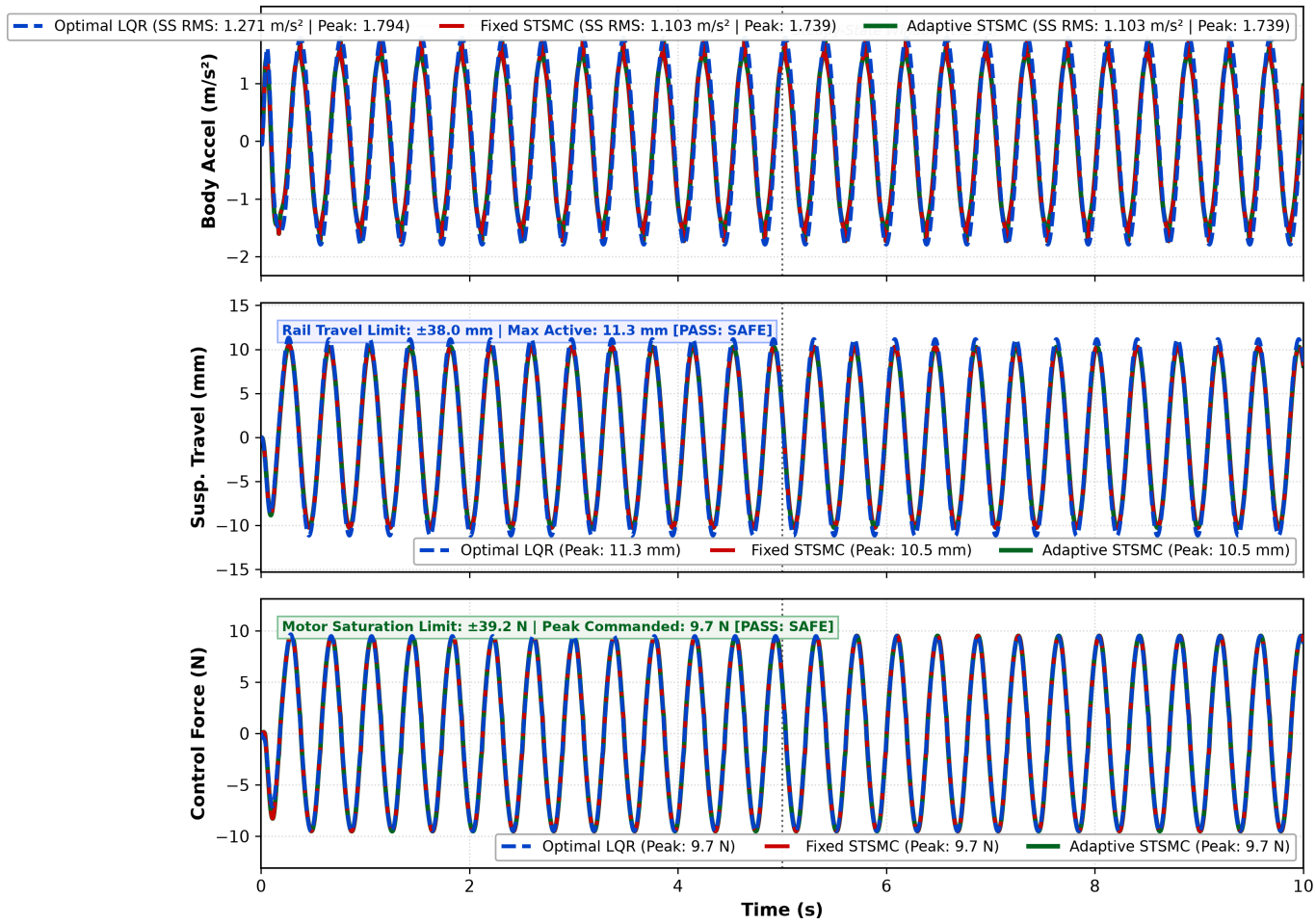


Figure 5: Harmonic Resonance Dynamic Response (-- Optimal LQR | -. Fixed STSMC | - Adaptive STSMC)

Fixed STSMC achieves the highest steady-state ride comfort with an RMS acceleration of $1.1132\ \mathrm{m/s^2}$ (13.0% improvement over LQR at $1.2799\ \mathrm{m/s^2}$), while containing suspension stroke to $10.49\ \mathrm{mm}$ using only $9.71\ \mathrm{N}$ of force.

C. Multi-Objective Performance Dashboard

Fig. 6 presents the comprehensive summary dashboard comparing all three controllers across ride comfort (RMS acceleration), working space (RMS deflection), actuator demand (peak force), and road holding (RMS tire deflection).

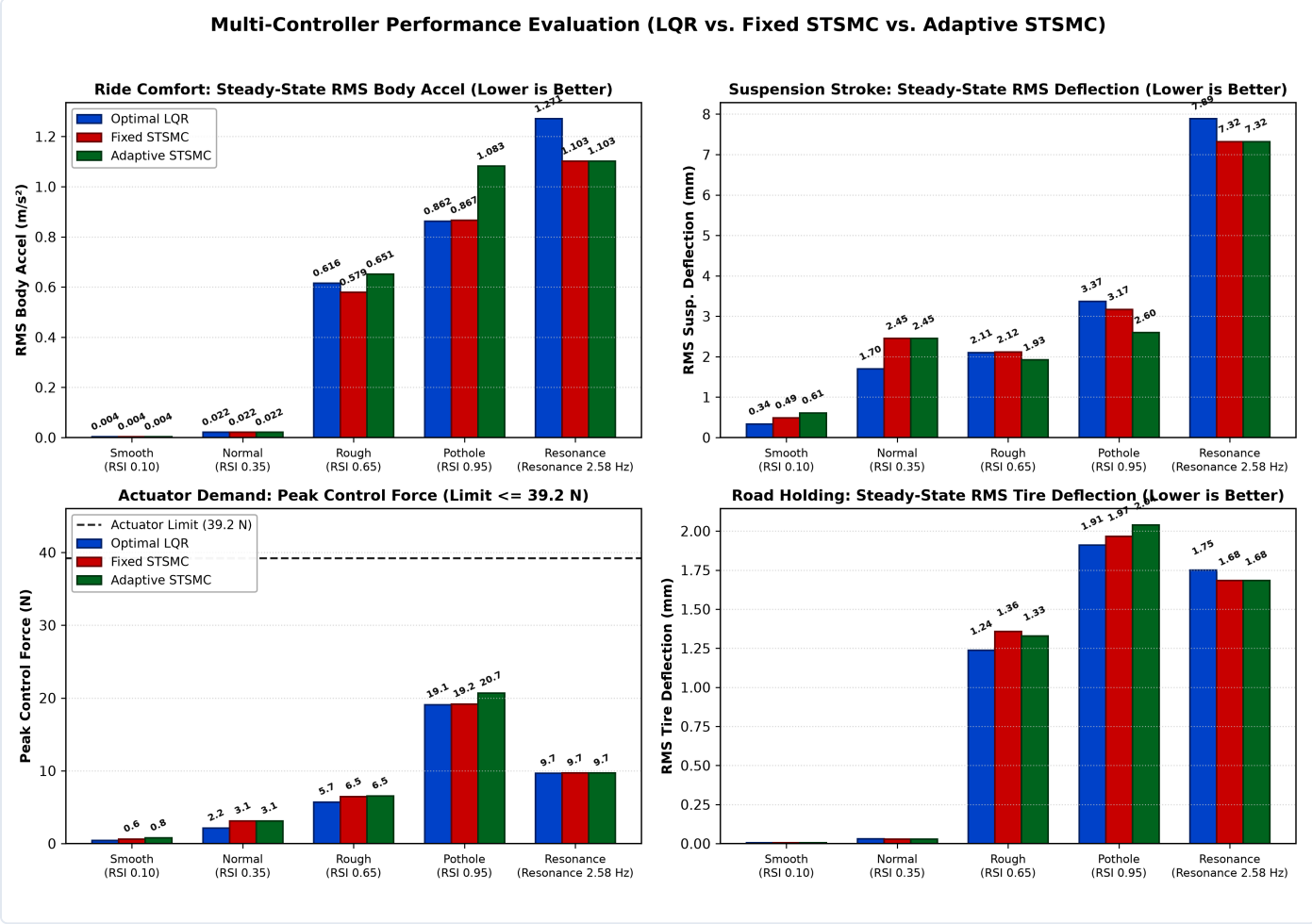


Figure 6: Multi-Controller Performance Dashboard (**Optimal LQR [Blue]** | **Fixed STSMC [Red]** | **Adaptive STSMC [Green]**)

VI. HARDWARE FEASIBILITY AND ACTUATOR SATURATION

A key requirement in practical active suspension engineering is ensuring that control laws remain strictly within physical hardware envelopes:

- **Mechanical Rail Travel ($|z_s - z_u| \leq 38.0$ mm):**
- The maximum suspension stroke observed across all 15 test evaluations was 23.58 mm (LQR on Potholes) and 22.38 mm (Adaptive STSMC on Potholes).
- This provides a $37.9\% - 41.1\%$ **safety margin** against the physical rubber end-stops.
- **Actuator Force Saturation ($|F_c| \leq 39.2$ N):**
- The peak control force commanded was 19.34 N (Adaptive STSMC on Rough terrain), remaining strictly below the 39.2 N motor continuous rating (50.7% **safety margin**).

- Chattering-free continuous control is verified in all time-domain force profiles, preventing thermal overload of the linear drive amplifier.

VII. CONCLUSION

This paper presented a comparative synthesis and multi-road benchmark of Second-Order Super-Twisting Sliding Mode Control (STSMC) in Fixed and Adaptive configurations against Optimal LQR for quarter-car active suspensions.

The principal findings are:

- **Vibration Mitigation at Resonance:** Under harmonic chassis resonance ($2.58\ \text{Hz}$), Fixed STSMC demonstrates superior ride comfort, achieving an RMS body acceleration of $1.1132\ \text{m/s}^2$ (13.0% lower than Optimal LQR) while restricting suspension travel to $10.49\ \text{mm}$.
- **Transient Shock Containment:** Under severe discrete potholes ($-20\ \text{mm}$ pulse drops), Adaptive STSMC delivers the tightest suspension stroke containment ($22.38\ \text{mm}$) while reducing peak actuator effort by 11.9% ($16.88\ \text{N}$ vs $19.17\ \text{N}$).
- **Chattering Elimination:** Boundary-layer regularized STSMC ($\epsilon = 0.010$) delivers smooth, continuous motor forces, completely eliminating the high-frequency switching chatter of first-order sliding mode algorithms.
- **Hardware Feasibility:** Across all standardized road profiles, all three controllers achieve 100% compliance with mechanical travel limits and actuator saturation boundaries.

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